

Pathophysiology and Health Impacts of Airborne Pollutants

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Topic: Physiological Effects of PM2.5, PM10, NO2, O3, and SO2

Pollutant Pathophysiology:

1. PM2.5 (Fine Particles): Inhaled into pulmonary alveoli; translocates to bloodstream; triggers systemic oxidative stress.
2. PM10 (Coarse Particles): Deposited in upper airways; causes throat irritation, coughing, and bronchitis exacerbation.
3. Nitrogen Dioxide (NO2): Causes bronchial airway inflammation, asthma exacerbation, and reduced lung capacity.
4. Ground-Level Ozone (O3): Powerful oxidant that damages lung lining tissue, causing chest tightness and reduced lung function.
5. Sulfur Dioxide (SO2): Induces bronchoconstriction in asthmatic individuals within minutes of exposure.